

working memory (KB)

- attention scores ($HL^2 \times 4 \text{ B}$, peak of one layer)
- KV cache ($2Ld_{\text{model}} \times 4 \text{ B} \times 6 \text{ layers}$)
- ours: 6-layer recurrent state

16 MB

4 MB

1 MB

256 KB

64 KB

16 KB

32

64

128

256

512

1024

context length L (tokens)

evaluated window: 32 tokens ($W = 64$ cycles, patch 2)
scores 16 KB are transient, but the KV cache is
96 KB and resident, twice our whole state

48 KB, flat in L

ours is the only line that is flat in L

